Lesson plan · for the teacher

Charts that tell the truth

Data science · Level 1 · Lesson 3

Time
40 min

Materials
Squared paper and coloured pencils; A small table of school data; A phone or tablet with Darsi, for the lab

Learning goals

- Choose bar, line, pie or scatter for a question.
- Spot a cut axis.
- Read a chart's title, axes and units.

Activity: Same data, four charts

1. Warm-up (5 min): Show the rainfall pie chart and ask what question it fails to answer.
2. Teach (8 min): Sketch a bar, a line, a pie and a scatter chart, each with the question it answers. Then show the cut-axis trick with the 70% and 74% bars.
3. Draw (15 min): Give every group the same table and a chart type. One group secretly draws a misleading version.
4. Gallery (7 min): Pin up the charts. Pupils vote on the clearest one and find the misleading chart, saying exactly what is wrong with it.
5. Wrap-up (5 min): Ask two pupils to say the finding in one sentence, with a number in it. Then give the exit ticket.

Discussion

- Which chart would you use for attendance over a term?
- Why do bars have to start at zero, but lines don't always?
- Where have you seen a misleading chart?

Exit ticket

- Draw a tiny honest bar chart of two numbers, with a title and labelled axes.

Try it in class tomorrow
Give groups the same small table and ask each for a different chart. Pin them up and ask which chart answers the question best.

Worksheet · for pupils

Charts that tell the truth

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Name _____ Class _____ Date _____

1 **A poster's bar chart has an axis starting at 90. What should you check first?**

◯ The colours ◯ Where the axis starts ◯ The title font

2 **Match each question to its chart.**

Which fruit does our class like best?	• • • •	Scatter chart
How did rainfall change this year?	• • • •	Pie chart
How is the school budget shared?	• • • •	Line chart
Do pupils who read more spell better?	• • • •	Bar chart

3 **This program draws a text bar chart of the pupils' favourite fruit. What is the second line it prints?**

```
fruits = ["Mango", "Banana", "Guava"]
votes = [5, 3, 2]
for i in range(3):
    print(fruits[i], "#" * votes[i])
```

◯ Banana ### ◯ Banana 3 ◯ Banana # ◯ Mango #####

4 **A bar chart whose axis starts at 68 shows a rise from 70% to 74% honestly.**

◯ True ◯ False

5 **Which should you check before believing a chart? Choose all that apply.**

☐ Where the bars start ☐ The title, axes and units

☐ Whether some months or groups were left out ☐ How bright the colours are

✎ **Think of a chart you saw recently in a newspaper, on TV or on a poster. Was it honest? How could you tell?**